# Leonardo KAWAKAMI PACHECO

School of Computing, Institute of Science Tokyo
Tokyo, Japan

email: leonardovpacheco@gmail.com
site: https://www.leonardopacheco.xyz

## Current Position

11/2024– School of Computing, Institute of Science Tokyo, Japan
JSPS Postdoctoral Fellow
KAKENHI Grant 24KF0214

## Previous Position

05/2023–10/2024 Institute of Discrete Mathematics and Geometry, TU Wien, Austria
Project Assistant under supervision of Juan P. Aguilera

## Education

2020–2023 Ph.D in Mathematics at Tohoku University, Japan
Thesis: Exploring the difference hierarchies on $\mu$-calculus and arithmetic — from the point of view of Gale–Stewart games
Advisor: Keita Yokoyama

2018–2020 M.Sc. in Mathematics at Tohoku University, Japan
Thesis: On the weak hierarchy of $\mu$-calculus
Advisor: Kazuyuki Tanaka

2014–2017 Undergraduate in Mathematics at Universidade Federal de Goiás, Brazil

2012–2013 Undergraduate in Computer Science (incomplete) at Universidade Federal de Goiás, Brazil

## Language Proficiency

English — Fluent (TOEIC Listening & Reading, 985 points)
Japanese — Fluent (Japanese Language Proficiency Test N1)
Portuguese — Native

## Awards

2023 Kawai Doctoral Thesis Award from Tohoku University

## Teaching

2021 Teaching Assistant — 数学基礎論特論 (Mathematical Logic — Graduate)
2021 Teaching Assistant — 計算機数学 A (Computability A)
2022 Teaching Assistant — 計算機数学 B (Computability B)

## Service

2026 8th Intuitionistic Modal Logic and Applications Workshop (IMLA'26) — Program Comitee Member
2026 Fuzzy Sets and Systems — Peer-reviewer
2026 Logic Journal of the IGPL — Peer-reviewer
2026 International Joint Conference on Automated Reasoning (IJCAR 2026) — Peer-reviewer
2026 Advances in Modal Logic (AiML 2026) — Peer-reviewer
2026 ACM/IEEE Symposium on Logic in Computer Science (LICS 2026) — Peer-reviewer
2025 Annals of Pure and Applied Logic — Peer-reviewer
2025 Theory and Applications of Craig Interpolation — Chapter Peer-reviewer
2025 International Conference on Automated Reasoning with Analytic Tableaux and Related Methods (TABLEAUX 2025) — Peer-reviewer

## Peer-Reviewed Publications

J.P. Aguilera, L. Pacheco, *Intuitionistic Gödel–Löb without sharps*, ACM Transactions on Computational Logic, Volume 26 (4), 1–14, 2025. DOI: 10.1145/3748649.

L. Pacheco, *The $\mu$-calculus' Alternation Hierarchy is Strict over Non-Trivial Fusion Logics*, Electronic Proceedings in Theoretical Computer Science, Volume 435, 93–103, 2025. DOI:10.4204/EPTCS.435.8.

J.P. Aguilera, R. Lubarsky, L. Pacheco, *Higher-Order Feedback Computation*, Lecture Notes in Computer Science, Volume 14773, 298–310, 2024. DOI:10.1007/978-3-031-64309-5_24.

L. Pacheco, K. Tanaka, *The alternation hierarchy of the mu-calculus over weakly transitive frames*, Lecture Notes in Computer Science, Volume 12468, 207–220, 2022. DOI:10.1007/978-3-031-15298-6_13.

## Preprints

L. Pacheco, *The Constructive $\mu$-calculus: Game Semantics and Non-Wellfounded Proof Systems*, arXiv:2604.23273

J.P. Aguilera, D. Fernández-Duque, L. Pacheco, *Polytopological Semantics for Intuitionistic Modal Logics*, arXiv:2604.23234

L. Pacheco, *Collapsing Constructive and Intuitionistic Modal Logics*, arXiv:2408.16428.

L. Pacheco, K. Yokoyama, *Determinacy and reflection principles in second-order arithmetic*, arXiv:2209.04082.

## Other Publications

L. Pacheco, W. Li, K. Tanaka, *On one-variable fragments of modal $\mu$-calculus*, Proceedings of CTFM 2019, World Scientific Publications, 17–45, 2022.

L. Pacheco, *Recent Results on Reflection Principles in Second-Order Arithmetic*, RIMS Kôkyûroku No.2228, 73–77, 2023.

L. Pacheco, *Epistemic possibility in Artemov and Protopopescu's intuitionistic epistemic logic*, RIMS Kôkyûroku No.2293, 66–71, 2024.

W. Li, L. Pacheco, K. Tanaka, *A fine hierarchy for the alternation-free modal $\mu$-calculus*, RIMS Kôkyûroku No.2346, 89–101, 2026.

Y. Nishimura, L. Pacheco, *On the Complexity of the SAT problem for the Hybrid Logic*, RIMS Kôkyûroku No.2346, 102–109, 2026.

## Invited Talks

2026 *A short overview of non-classical modal logics* at Symposium on Advances in Mathematical Logic 2026.

2025 *On diamonds in constructive modal logic* at Conference on Non-Classical Modal Logics, National Taiwan University.

## Selected Talks

2026 *A non-wellfounded proof system for the constructive $\mu$-calculus* at Ookayama Workshop on Modal Logic.

2026 *Decidability via fully-labeled non-wellfounded proofs* at LLAL@GSIS (XIII).

2026 *Gödel–Dummett meets Gödel–Löb* at Arithmetical and Modal mu-Calculi: Recent and Future Advances.

2025 *Epistemic possibility in Artemov and Protopopescu's intuitionistic epistemic logic* at 18th Asian Logic Conference

2025 IGL *via $\omega$-rules* at Logic Colloquium 2025.

2025 *Topological Semantics for* IS4 at LLAL@GSIS (VI).

2024 *Collapsing Constructive and Intuitionistic Modal Logics* at Australasian Association for Logic 2024 Conference.

2024 IGL *without sharps* at Wormshop 2024.

2024 *Higher-order feedback computation* at Computability in Europe 2024.

2024 *A constructive variation of* GL at MSJ Spring Meeting 2024.

2024 *The $\mu$-calculus' Alternation Hierarchy is Strict over Non-Trivial Fusion Logics* at Fixed Points in Computer Science 2024.

2023 *Game Semantics for the Constructive $\mu$-calculus* at 15th Latin American Workshop on New Methods of Reasoning.

2023 *Fixed-points in epistemic logic* at MSJ Spring Meeting 2023.

2022 *The alternation hierarchy of the $\mu$-calculus over weakly transitive frames* at 28th Workshop on Logic, Language, Information and Computation.

2022 *Determinacy and reflection principles in second-order arithmetic* at Workshop on Reverse Mathematics and its Philosophy, joint presentation with Keita Yokoyama.

2022 *On the degrees of ignorance: via epistemic logic and $\mu$-Calculus* at SOCREAL 2022.